

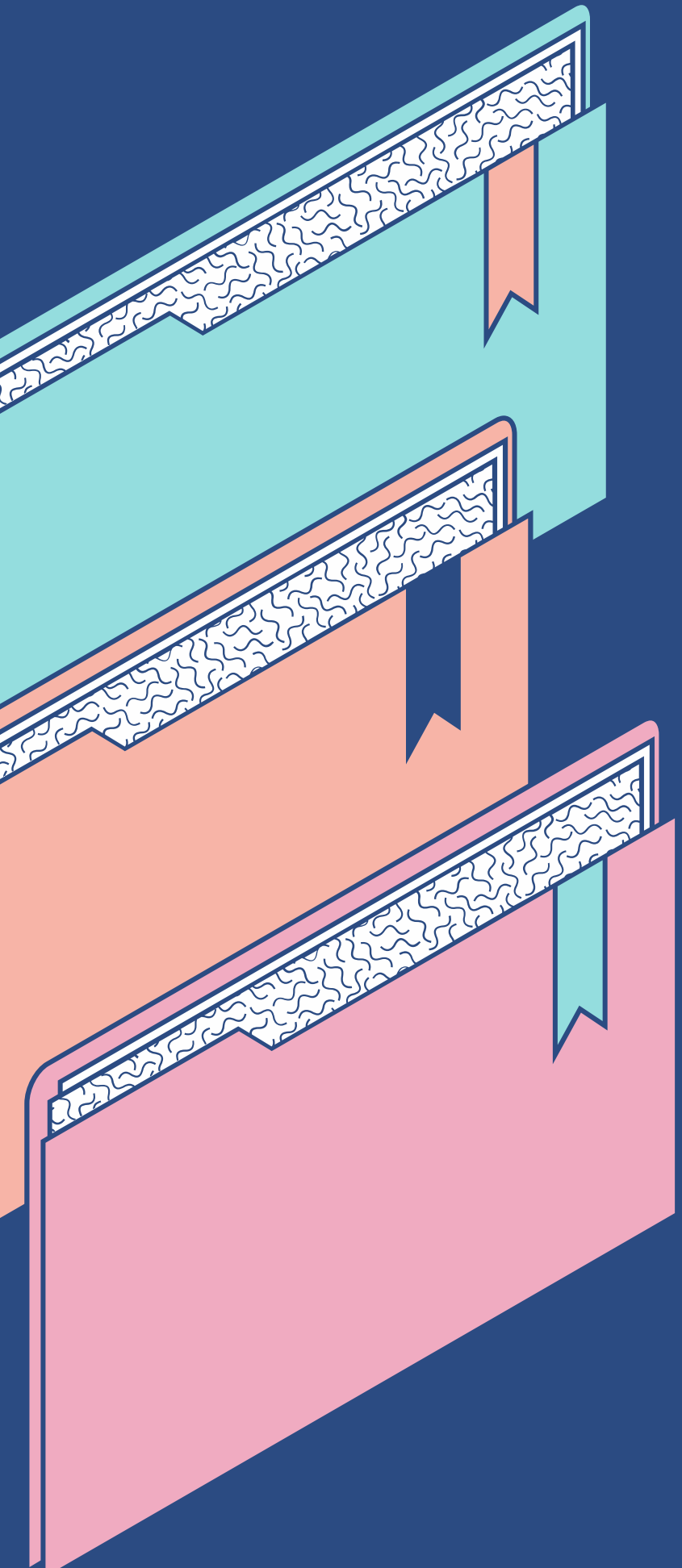


Group 4

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Joint BERT Span-Based Model for Aspect-Based Sentiment Analysis (ABSA)

- Using MAMS Dataset
- Task: End-to-end ABSA — ATE (Aspect Term Extraction) + ASC (Aspect Sentiment Classification)
- Model: Joint BERT with CRF for ATE and advanced span representation for ASC
- Dataset: MAMS – Multi-Aspect, Multi-Sentiment
- Key Innovation: Span-based ASC with start + end + mean pooling

Dataset: MAMS (Multi-Aspect Multi-Sentiment)

Why MAMS?

- Standard ABSA datasets (SemEval 2014–2016) often have few aspects per sentence and consistent sentiment.
- MAMS is designed for challenging, realistic scenarios:
- Each sentence has ≥ 2 aspects with different sentiments.
- Forces models to learn fine-grained context modeling.

MẪU 1:

Câu gốc: the decor is not special at all but their food and amazing prices make up for it .

Số từ: 18 | Số aspect: 3

	Aspect Term	From	To	Polarity
0	decor	1	2	negative
1	food	9	10	positive
2	prices	12	13	positive

the **decor** is not special at all but their **food** and amazing **prices** make up for it .

POSITIVE **NEUTRAL** **NEGATIVE**

THỐNG KÊ TOÀN BỘ TRAIN SET (n = 4297):

- Trung bình aspect: 2.60
- Max aspect: 11
- Phân bố polarity:
 - POSITIVE: 3380 (30.2%)
 - NEUTRAL : 5042 (45.1%)
 - NEGATIVE: 2764 (24.7%)

Preprocessing

Goal: Convert raw JSON → BERT-ready tensors with token-level (ATE) and span-level (ASC) labels.

Key Steps (create_bert_samples)

BERT Tokenization (WordPiece)

Input: ["The", "food", ...] → [CLS] The food ... [SEP]

Build word_map: maps subword → original word index

ATE Labels (B-I-O)

B-ASP: first subword of aspect

I-ASP: subsequent subwords

O: non-aspect

ASC Labels (Span-based)

span_indices: [MAX_ASPECTS, 2] → [[start, end], ...]

span_sentiments: [0, 1, 2, -100, ...] (ignore padding)

span_mask: [1, 1, 0, ...] (valid vs padding)

text

Input: "New York steak"

→ Tokens: [New, York, steak]

→ ATE: [B-ASP, I-ASP, I-ASP]

→ Span: [[2, 4]] (subword indices)

Model Architecture

JointBERT_Span_ABSA

```
BERT → [Hidden States]
      ↳ ATE Head (CRF)
      ↳ ASC Head (Span Classifier)
```

Main Components:

BERT Backbone: Extracts hidden states from input.

1. ATE Head (Token-level):

Projection: Linear(768 → 3) to create emissions.

CRF: Decodes tags (B-I-O), handles dependencies (e.g., I-ASP after B-ASP).

2. ASC Head (Span-level):

Extract span rep: Concat start_rep + end_rep + mean_rep (768*3=2304 dims).

Projection: Linear(2304 → 3) to classify sentiment.

Dropout: 0.1 to prevent overfitting.

Forward Pass:

Always run ATE.

ASC only runs if span_indices provided (training/eval).

Loss: Joint loss = 0.5 * ASC_loss (CE) + 0.5 * ATE_loss (CRF negative log-likelihood).